

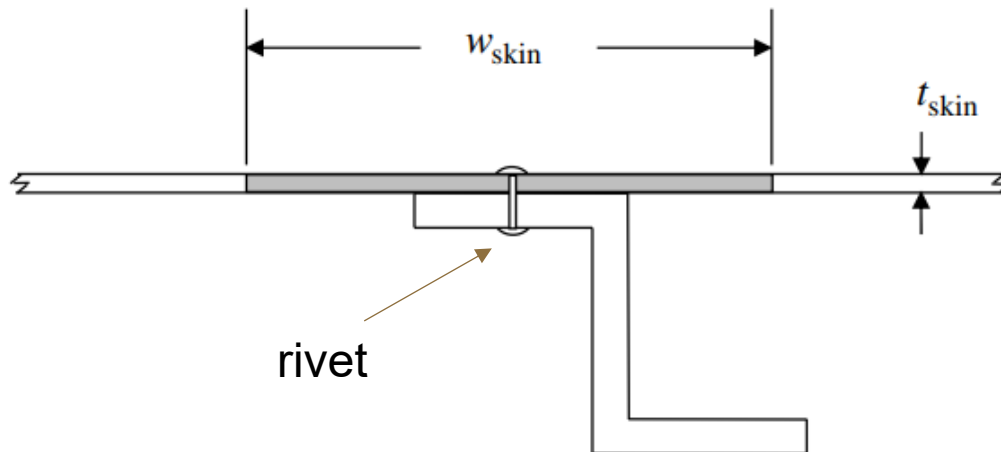
5.1: Idealization of Beam Cross-Sections

- So how much web cross-sectional area should be included with the actual stringer cross-sectional area to form the effective stringer area?
- For portions of the wing under **compression**, the web material is less effective than the stringers in resisting the compressive load
- However, some web material immediately adjacent to each stiffener is stiffened (i.e. strengthened) somewhat due to its attachment to the stiffener
- The **“30x Thickness” Rule** is commonly used to estimate this amount of web material, which is then “lumped” with the stiffener to form the effective stringer area → the “fore to aft” width would then be $w_{skin} = 30 * t_{skin}$
- From a cross-sectional point of view, this amount of width is assumed to be distributed equally on either side of the rivet (i.e. the point of attachment between the stringer and skin) → $15 * t_{skin}$ on each side of the rivet centerline

5.1: Idealization of Beam Cross-Sections

- The figure below shows the cross-section of a z-beam stiffener attached to a skin by a single line of rivets
- In this case, the effective stringer cross-sectional area would be:

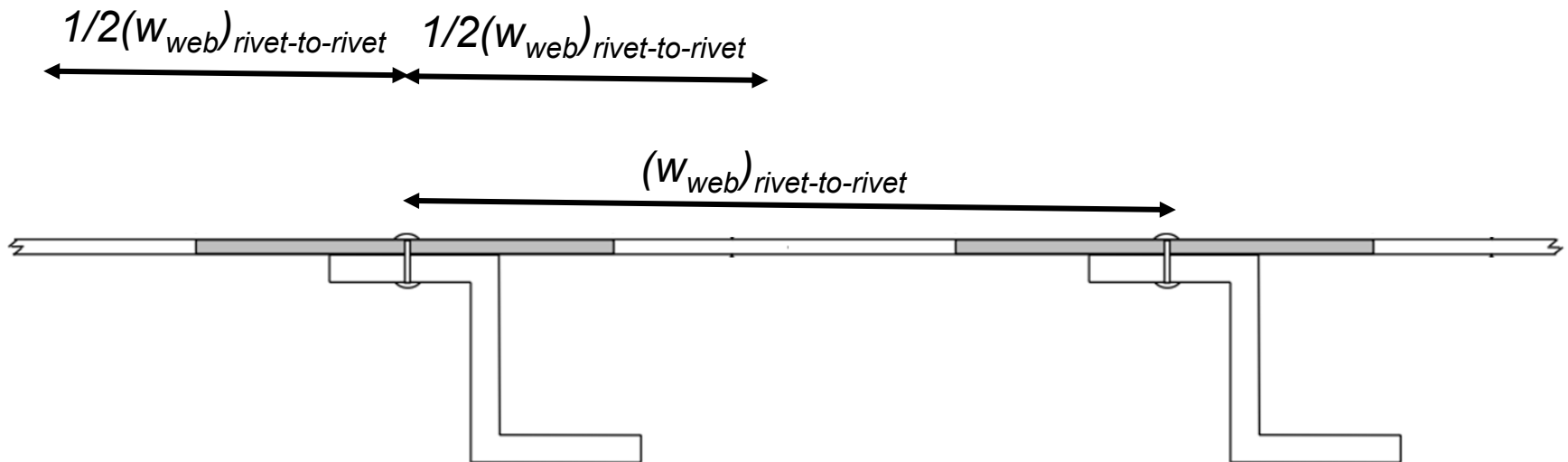
$$A_{\text{eff}} = A_{\text{z-beam}} + \overbrace{\left[30(t_{\text{skin}}) \right]}^{w_{\text{skin}}} t_{\text{skin}}$$



5.1: Idealization of Beam Cross-Sections

- For portions of the wing in **tension**, all of the web material between successive stiffeners is considered effective in resisting the tensile load; so in this case:

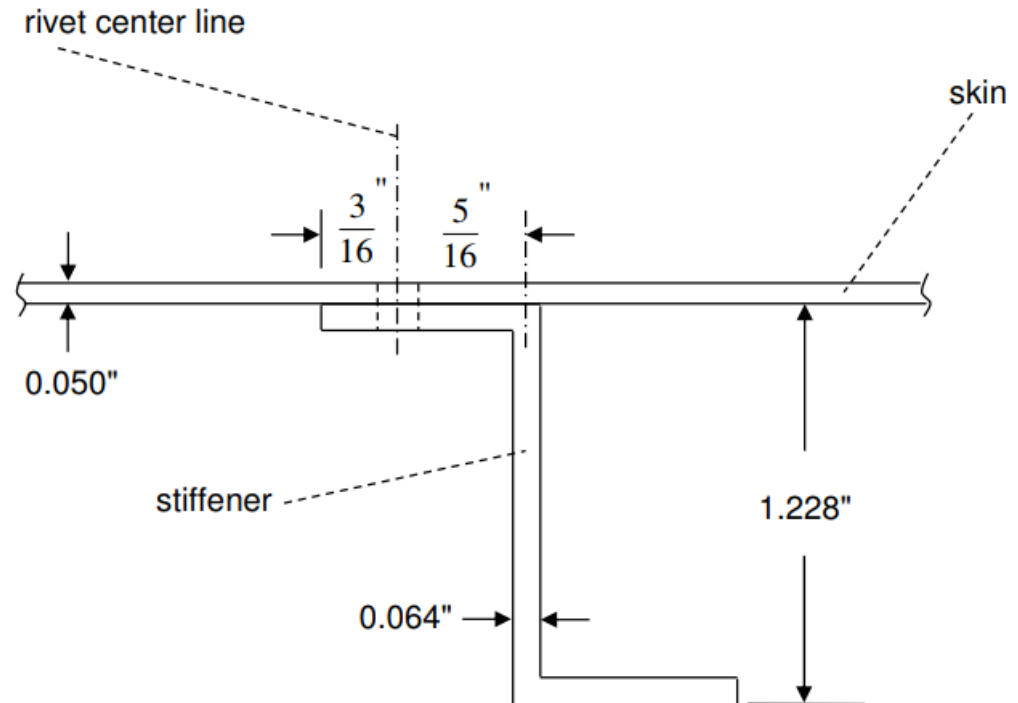
$$A_{\text{eff}} = A_{\text{z-beam}} + \left[\frac{1}{2} (A_{\text{web}})_{\text{Rivet-to-Rivet}} \right]_{\text{Left Side of Rivet}} + \left[\frac{1}{2} (A_{\text{web}})_{\text{Rivet-to-Rivet}} \right]_{\text{Right Side of Rivet}}$$



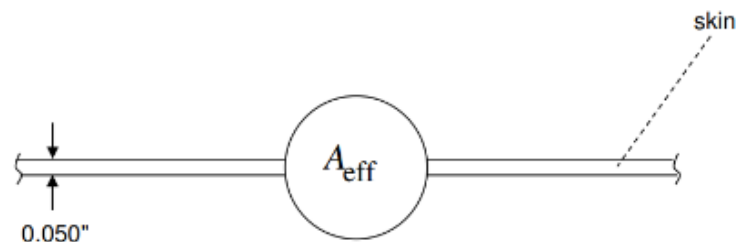
5.1: Idealization of Beam Cross-Sections

Example 5.1.1

Idealize the structure consisting of a z-beam stiffener attached to a skin by a single row of rivets. The skin and stringer are under compression.



The idealized structure will appear as shown here → we need to find A_{eff}



5.1: Idealization of Beam Cross-Sections

- So $w_{skin} = 30 * t_{skin} = 30 * (0.05) = 1.5''$
- The idealization involves “lumping” half of the 1.5" (i.e. 0.75”) from each side of the rivet centerline, thus:

$$\underline{A_{eff}}$$

$$A_{eff} = A_{stringer} + A_{skin}$$

$$= \overbrace{\left[(1.228'') (0.064'') + 2 (0.468'') (0.064'') \right]}^{0.1385 \text{ in}^2} + \overbrace{(1.5'') (0.050'')}^{0.075 \text{ in}^2}$$

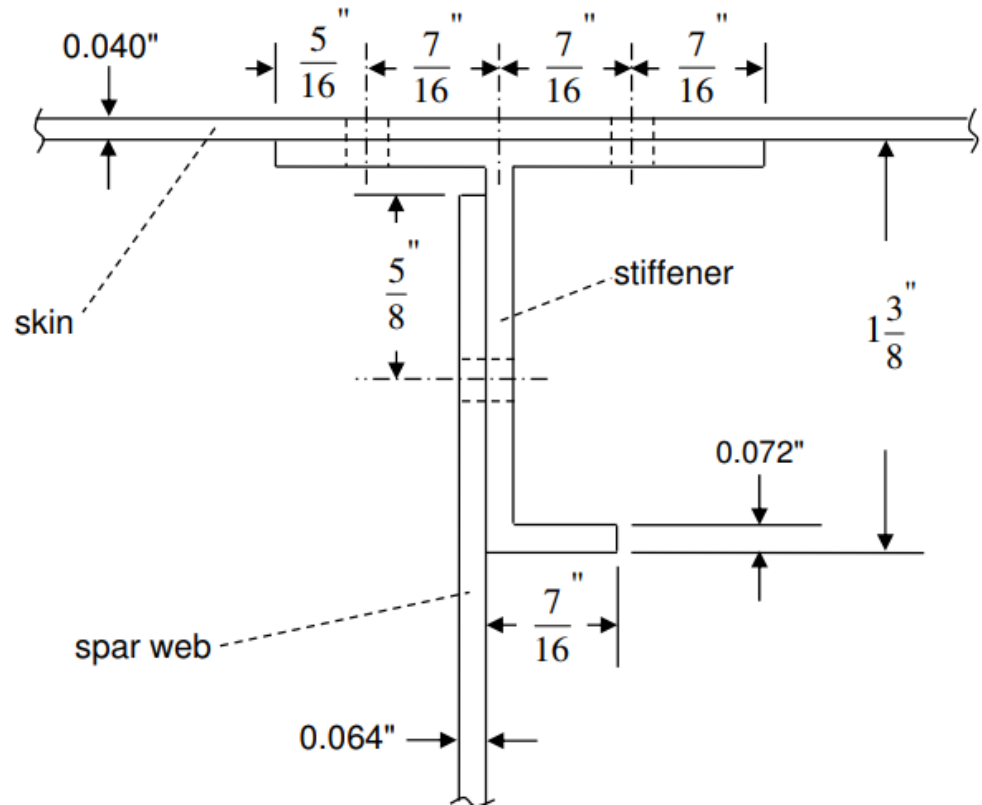
$$A_{eff} = 0.2135 \text{ in}^2$$



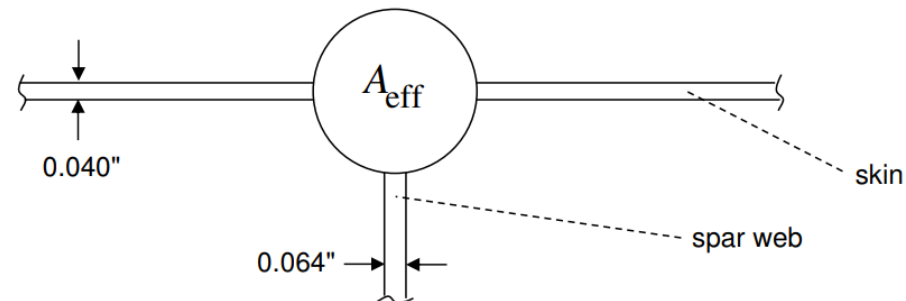
5.1: Idealization of Beam Cross-Sections

Example 5.1.2

Idealize the upper spar cap consisting of a J-beam stiffener attached to the skin (by a double row of rivets) and attached to the vertical spar web (by a single row of rivets). All are under compression.



The idealized structure will appear as shown here → we need to find A_{eff}



5.1: Idealization of Beam Cross-Sections

- Let's examine w_{skin} for each rivet
- Using the “30x Thickness” Rule, we have $w_{skin} = 30 * t_{skin} = 30 * (0.04) = 1.2$ ”
- Normally the idealization would involve “lumping” half of the 1.2” (i.e. 0.6”) from each side of the rivet centerline; however, the distance between the two rivet centerlines is $(7/16)'' + (7/16)'' = 0.875$ ”
- Because $0.875 < 1.2$, we modify our idealization as follows: $w_{skin} = 0.6'' + (7/16)'' + (7/16)'' + 0.6'' = 2.075$ ”



5.1: Idealization of Beam Cross-Sections

- Let's now examine the vertical spar web (i.e. w_{web})
- Using the “30x Thickness” Rule, we have $w_{web} = 30 * t_{web} = 30 * (0.064) = 1.92''$
- Normally the idealization would involve “lumping” half of the 1.92" (i.e. 0.96") from each side of the rivet centerline; however, the upper end of the spar webbing is only $(5/8)'' = 0.625''$
- So we modify our idealization as follows: $w_{skin} = (5/8)'' + 0.96'' = 1.585''$

$$A_{eff} = A_{stringer} + A_{skin} + A_{web}$$

- Effective area is then:

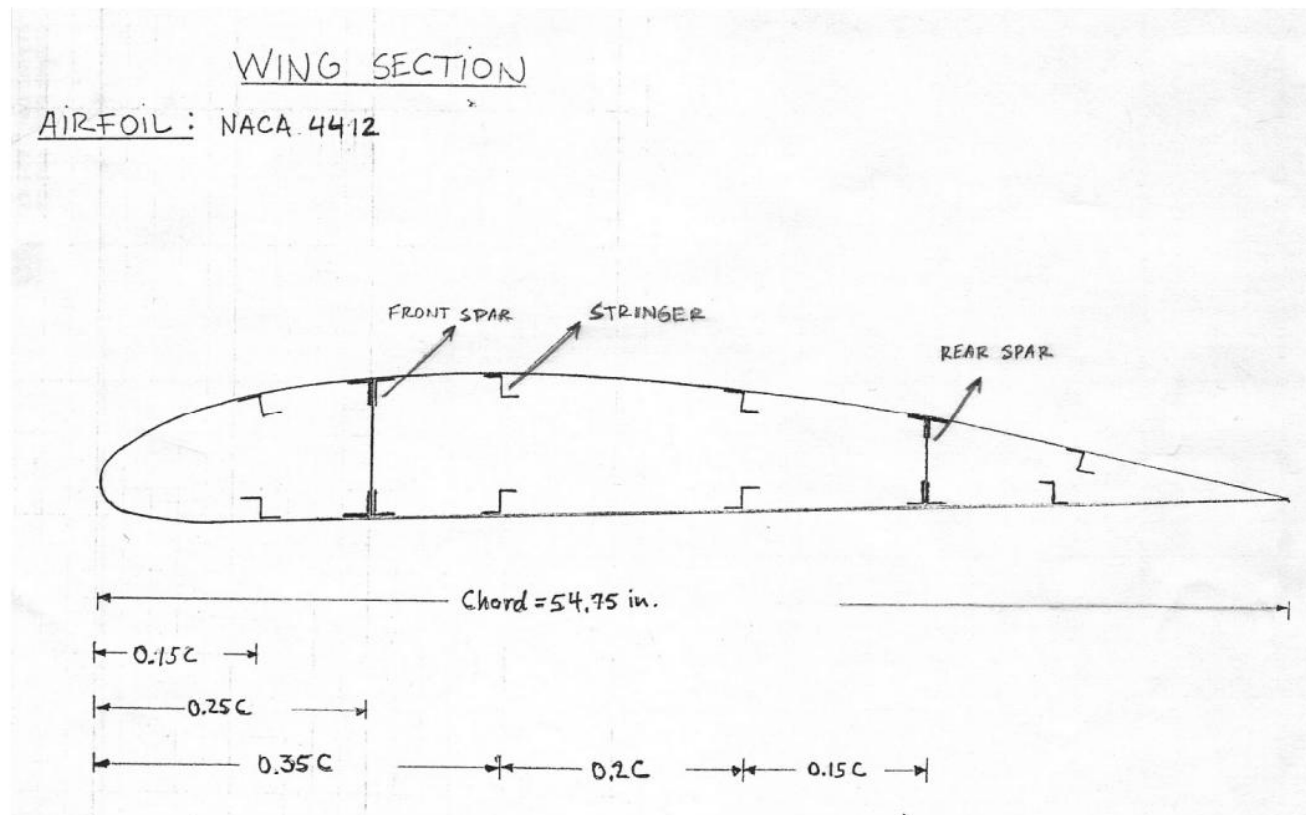
$$= \left\{ \begin{array}{l} \overbrace{\left[(1.625'')(0.072'') + (1.231'')(0.072'') + (0.4375'')(0.072'') \right]}^{0.2371 \text{ in}^2} \\ \underbrace{+ (2.075'')(0.040'')}_{0.083 \text{ in}^2} + \underbrace{(1.585'')(0.064'')}_{0.1014 \text{ in}^2} \end{array} \right\}$$

$$A_{eff} = 0.4215 \text{ in}^2$$



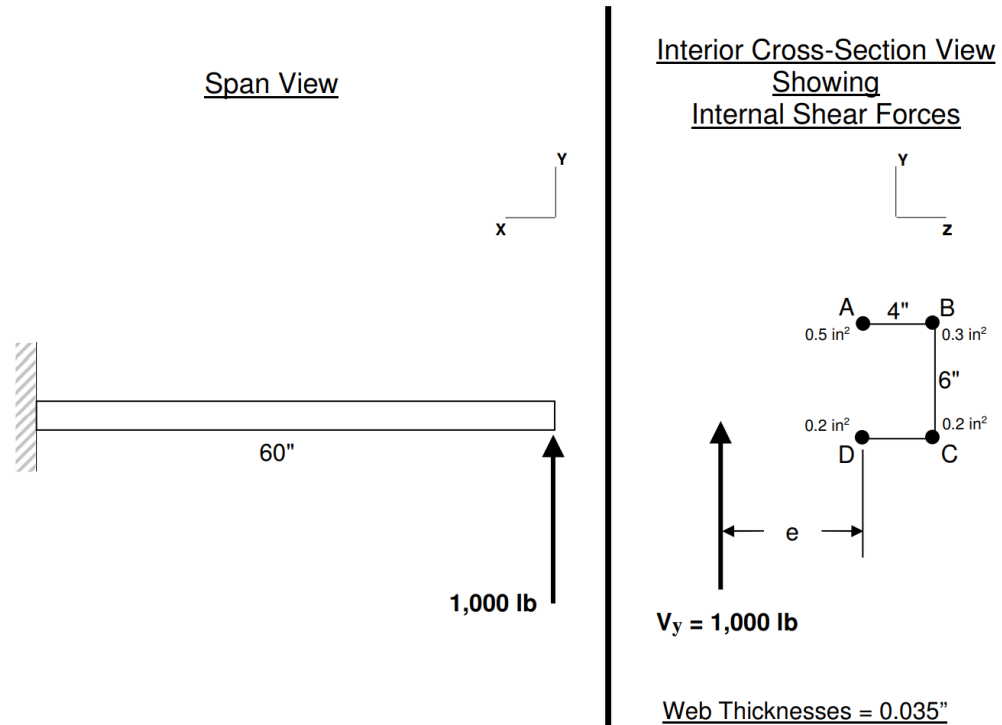
5.1: Idealization of Beam Cross-Sections

- Here's a look at Problem 5.1 in the textbook (to be included in a HW next week):
 - The wing below must be idealized in order to perform structural analysis; i.e., effective stringer areas must be determined. Assume that the major portion of the load will be carried by the center cell [from spar to spar]. Idealize this cell for analysis, assuming that both upper and lower portions will have to take compression at some time during its loading history.



5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- Consider the cantilevered beam below. It has a single transverse load applied at the free end. The cross-section has already been idealized and is ready for analysis. This means that the stringer areas are effective stringer areas, where the stringers contain σ and the webs contain q :

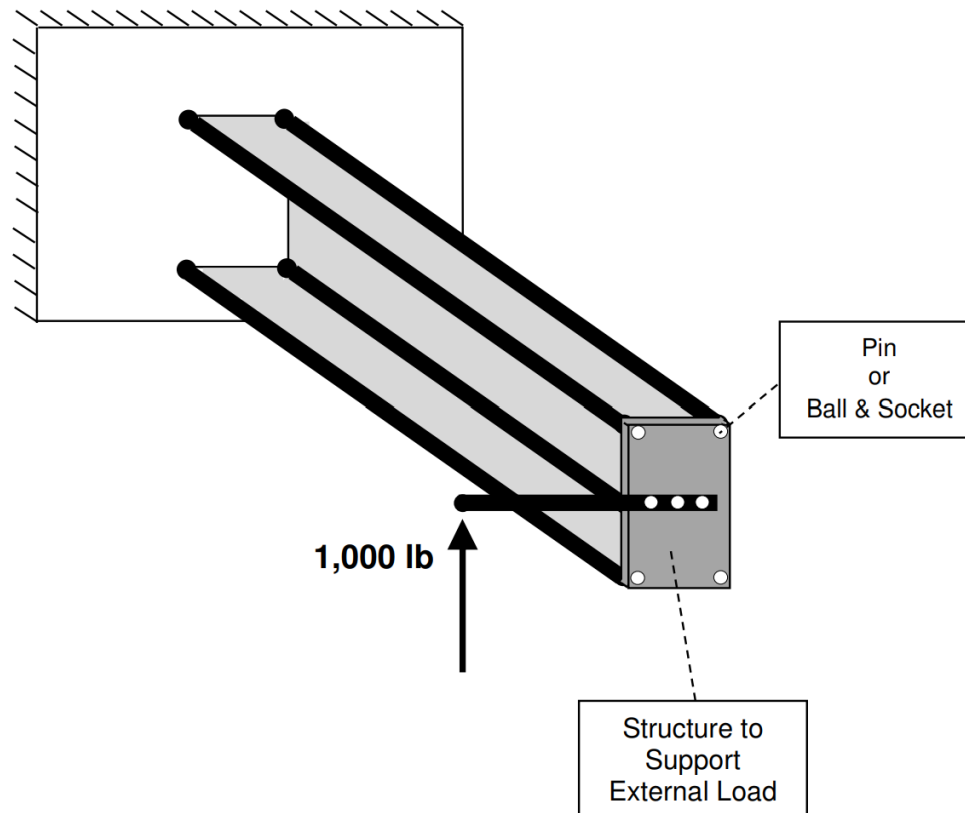


Find

- the normal stress in each stringer at the wall;
- the shear flows and show your answers on a diagram;
- the distance "e" (indicating the location where the 1,000 lb load must act) so that the beam will experience no torsion and show your answer on a diagram;
- the location of the Shear Center and show your answer on a diagram.

5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- The cross-section view shown on the previous slide is not the external free end, but rather an interior face
- An actual view of the entire beam is shown below, including a structure at the end to support external loads



5.2: Idealized Cross-Section Beams Under Transverse Loading: Straight Webs / Open Cross-Sections

- The cross-section view with the structure at the end removed looks like this:

